



**RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF TECHNOLOGY**

**Bachelor of Information and Communication Technology
First Year - Semester I Examination – September/October 2022**

ICT 1305 – PROGRAM DESIGN AND PROGRAMMING

Time: Three (03) hours

Answer all questions from Part A.

Answer any Three (03) Questions from Part B.

Part A

1.

- a) What are the four (04) types a function can be implemented? [4 marks]
b) Draw a relevant flowchart to the given algorithm.

START

1. *PROMPT* the user to enter an integer

2. *READ* the integer, num

3. *EXTRACT* the first least significant number, $firstDigit = (num \% 10)$

4. *EXTRACT* the second least significant number, $secondDigit = (num \% 10) \% 10$

5. *COMPUTE* for the sum of the two least significant numbers in the integer,
 $sum = firstDigit + secondDigit$

6. *WRITE* the sum of the two least significant numbers in the integer, sum

END

[8 marks]

- c) Write a C program that reads three (03) integer values and check if it is possible to make a triangle with them. Also calculate the perimeter of the triangle if the above said values are valid. [10 marks]
d) The following function is implemented to calculate the sum of natural numbers using recursion. Fill in the blanks and rewrite the function.

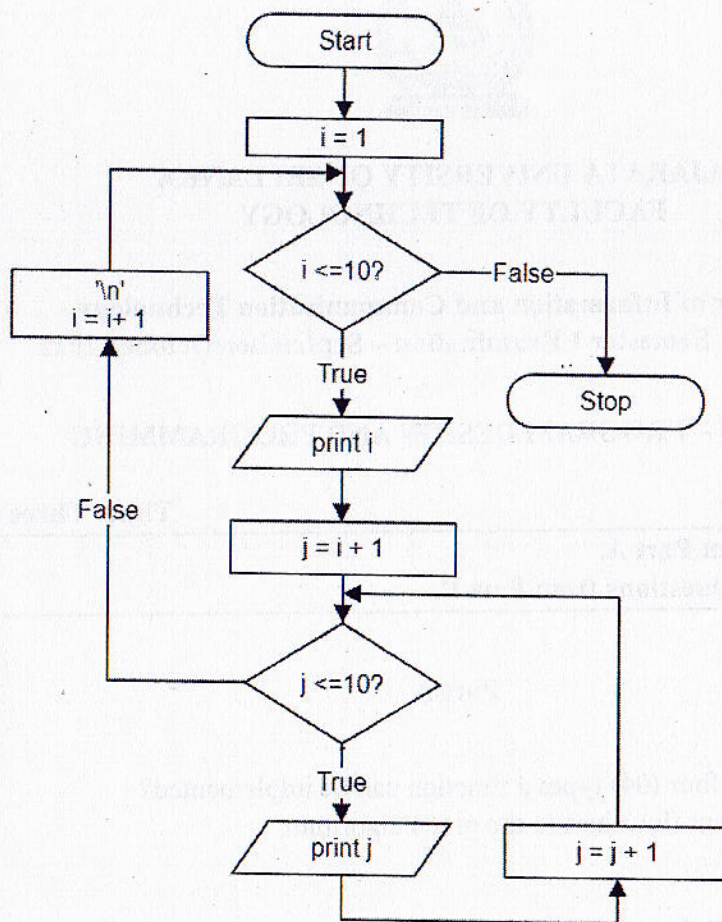
```

_____ sum(int n) {
    if (n != _____)
        return _____ + _____(n-1);
    _____
    return n;
}

```

[5 marks]

e) Briefly explain the output of the following flowchart.



f) "Following a recipe is a type of an algorithm." Do you agree with the above statement? Justify your answer. [8 marks]

[5 marks]

[Total 40 marks]

Part B

2.

- a) What are four (04) rules for generating variable names in pseudocode? [5 marks]
 b) Draw a flowchart to calculate the area of a triangle, circle, square or rectangle based on the user's choice. [8 marks]
 c) Write a C program that takes an integer as input and returns product of digits of that number. Ex: Input: 154 Output: 20 ($1 \times 5 \times 4$)

Start

1. Get the number
2. Declare a variable to store the product and set it to 1
3. Repeat the next two steps till the number is not 0
4. Get the rightmost digit of the number with help of remainder '%' operator by dividing it with 10 and multiply it with product.
5. Divide the number by 10 with help of '/' operator
6. Print or return the product.

End

[5 marks]

- d) Write two (02) real world applications of 2D arrays.

[2 marks]

[Total 20 marks]

3.

- a) What are the advantages of using recursive functions? [3 marks]
 b) Write an algorithm or a pseudocode to calculate the wages of a laborer according to the following table.

TIME	WAGE
First 10 hrs.	Rs 60
Next 6 hrs.	Rs 15
Next 4 hrs.	Rs 18
Above 10 hrs.	Rs 25

[6 marks]

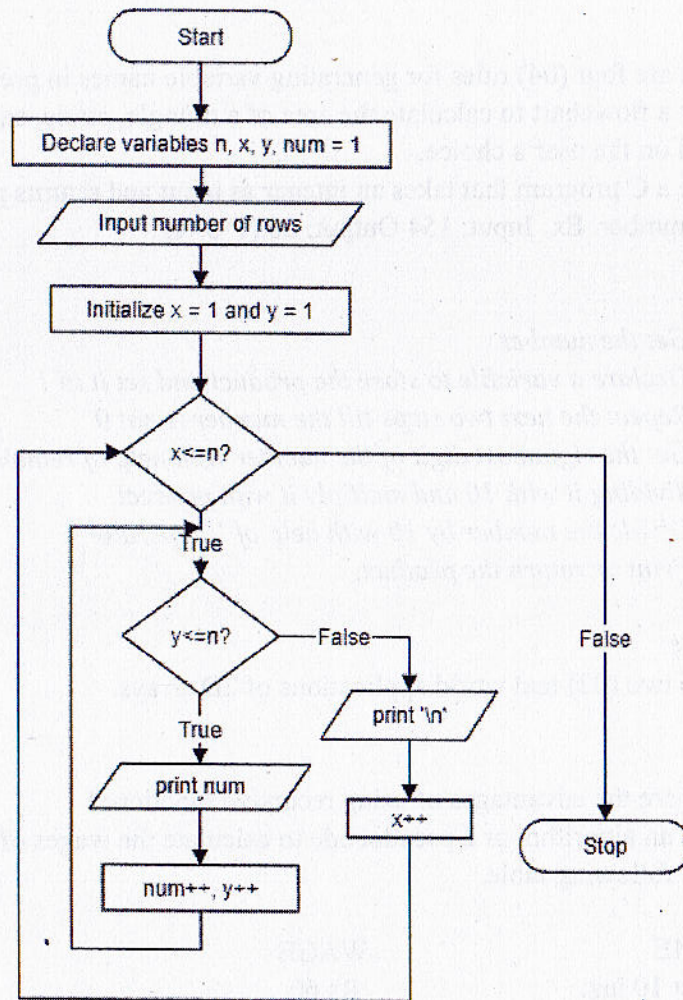
- c) Write a C program to print the following star pattern using loops.

```

* * * * *
 * * * * *
  * * * * *
   * * * * *
    * * * * *
     * * * *
      * * *
       * *
        *
  
```

[6 marks]

d) Write the output of the following flowchart.



[5 marks]

[Total 20 marks]

4.

a) Write five (05) real world uses of flowcharts.

[5 marks]

b) The following program is written to reverse a user given integer. Identify five (05) syntax errors in the given program and rewrite just the corrected code lines.

```

#include <stdio.h>

int main {
    int n, reverse = 0, remainder;

    printf("Enter an integer: ")
    scanf("%d", n);

    while (n != 0) {
        remainder = n % 10;

```

```
reverse = reverse * 10 + remainder;
n /= 10;
```

```
printf("Reversed number = %d", Reverse);
```

```
return 0;
```

```
}
```

[5 marks]

- c) What is a pointer in programming? Explain using a suitable example. [4 marks]
 d) Identify the purpose of the following C program. Justify your answer using a suitable input value.

```
#include <stdio.h>
int main() {
    int n, i;
    int x = 1;
    printf("Enter an integer: ");
    scanf("%d", &n);

    for (i = 1; i <= n; ++i) {
        x *= i;
    }
    printf("Answer is %i", x);
    return 0;
}
```

[6 marks]

[Total 20 marks]

5.

- a) Briefly explain the following C program line by line mentioning its output if n is given as 5.

```
#include <stdio.h>
int main() {
    int n, i;
    int f = 1;
    printf("Enter an integer: ");
    scanf("%d", &n);
    for (i = 1; i <= n; ++i) {
        f *= i;
    }
    printf("The answer is: %d, f);
    return 0;
}
```

[6 marks]

b) A cricket kit supplier sells bats, wickets & balls. Draw a flowchart to generate sales bill. Input the following from the console, price of each item & quantity of each item.

I. Calculate the total sale amount & add 17.5 % sales tax if the total sales amount >300000 & add 12.5 % if the total sales amount is >150000 & 7 % otherwise.

II. Display the total sales amount, the sales tax & the grand total.

[10 marks]

c) Write down four (04) good programming practices.

[4 marks]

[Total 20 marks]

--End--